



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



NASA Core Flight System Health and Safety Denial-of-Service Daily Review

Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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NASA Core Flight System Health and Safety Denial-of-Service Daily Review

1. Executive Summary

CISA ICS reporting identifies a NASA Core Flight System Health & Safety application vulnerability where processing a routine Housekeeping Telemetry request can crash the application through a segmentation fault. Aerospace and space-system defenders should assess cFS-derived deployments, lab replicas, and mission-software assurance workflows.

This report was selected for the 2026-07-31 UTC daily Lost Boys Cyber production run because the item is recent or materially relevant to defenders within the last 24-72 hours; where fewer fresh primary sources were available, the window was expanded to seven days and documented in the source review.

2. Intelligence Requirements

Question	Current Answer	Confidence
What happened?	For space and aerospace operators, availability faults in flight/mission frameworks matter because restart, patch, and validation cycles may be constrained. The practical defender focus is asset discovery, version verification, testbed reproduction under authorization, and compensating controls around telemetry-message handling paths.	High
Who should prioritize review?	Space systems, mission software, aerospace testbeds, satellite ground engineering	Medium
What is the immediate defender action?	Validate exposure, tune detections, run hunts, and confirm response ownership.	High

3. Source Review & Confidence

Source	Contribution	Confidence
CISA ICSA-26-197-03 NASA cFS HS advisory	Primary advisory describing cFS HS application crash condition.	High
CISA ICS advisories index	Official advisory corpus context.	High
NASA Core Flight System project	Project/source context for cFS ecosystem and deployments.	Medium

4. Threat Activity Overview

For space and aerospace operators, availability faults in flight/mission frameworks matter because restart, patch, and validation cycles may be constrained. The practical defender focus is asset discovery, version verification, testbed reproduction under authorization, and compensating controls around telemetry-message handling paths.

5. Affected Sectors and Exposure

Sector / Asset Group	Exposure Consideration	Analyst Note
Space systems, mission software, aerospace	Current reporting maps directly or	Prioritize assets with external exposure,

testbeds, satellite ground engineering	indirectly to this environment class.	privileged identity reach, or operational dependencies.
Third-party providers	Shared platforms and managed services can carry the operational impact.	Request patch, detection, and incident-notification evidence.
Remote workforce / identity plane	Many campaigns begin with credential, phishing, or trusted-channel abuse.	Review MFA, conditional access, impossible travel, and helpdesk reset controls.

6. Aviation and Aerospace Sector Impact

Segment	Operational Relevance	Likely Exposure Path	Defender Action
Space systems, mission software, aerospace testbeds, satellite ground engineering	Current source material has aviation/aerospace operational relevance and may affect safety-adjacent business continuity, regulated operations, or mission support.	Exposure may occur through internet-facing services, trusted suppliers, identity workflows, operational IT, or sector-specific communications.	Validate ownership, patch and monitor relevant systems, rehearse degraded operations, and confirm third-party incident notification paths.
Third-party and supply chain	Aviation outcomes often depend on shared service providers, airport/airline vendors, OEM/MRO integrations, and cloud/SaaS platforms.	Vendor compromise, unavailable managed platforms, delayed disclosure, or shared credentials can cascade beyond one organization.	Require vendor attestations, segment integrations, maintain offline continuity plans, and define escalation contacts before an incident.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed Technique	Evidence
Initial Access	Phishing, exploit public-facing application, or supply-chain trust abuse	Source reporting describes campaign or sector activity that begins through exposed services, trusted communications, or software dependencies.
Execution	Script, malware, or administrative-tool execution	Defender review should focus on process creation, browser-mediated execution, package scripts, or suspicious service-account activity.
Persistence / Defense Evasion	Valid accounts, relay infrastructure, tool renaming, or trusted process abuse	Activity may blend into normal administrative or browser/cloud traffic.
Impact	Data theft, operational disruption, ransomware, or mission degradation	Sector impact depends on operational criticality and third-party dependencies.

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
Source-reported domains, IPs, hashes, or package names	Observable	No hard IOC was safely extracted during automated run unless listed in source material; analysts should enrich source pages before blocking.

Abnormal sign-in, script execution, web-service child processes, package publish events	Behavioral signal	Preferred for durable detection when hard IOCs are absent or volatile.
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9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Suspicious command execution after trusted-app or browser activity	EDR process telemetry	Identify post-compromise execution hidden behind legitimate parent processes.
Unusual outbound traffic from servers or developer endpoints	DNS, proxy, firewall, EDR network events	Surface payload retrieval, C2, or data staging.
Identity anomalies following phishing or supply-chain exposure	Entra ID, Okta, VPN, SSO logs	Find compromised accounts before lateral movement.

9.2. Sigma / Detection Content

```

title: Daily Threat Review Suspicious Trusted Process Follow-On Activity
status: experimental
description: Detects suspicious command or scripting tools launched from browsers, collaboration
clients, package managers, or service parents during daily threat-review windows.
logsource:
  product: windows
  category: process_creation
detection:
  trusted_parent:
    ParentImage|contains:
      - chrome.exe
      - msedge.exe
      - teams.exe
      - npm.cmd
      - node.exe
      - w3wp.exe
      - outlook.exe
  suspicious_child:
    Image|endswith:
      - \powershell.exe
      - \pwsh.exe
      - \cmd.exe
      - \wscript.exe
      - \cscript.exe
      - \rundll32.exe
      - \regsvr32.exe
  condition: trusted_parent and suspicious_child
fields:
  - UtcTime
  - Computer
  - ParentImage
  - Image
  - CommandLine
level: medium

```

9.3. Hunt Query

```

DeviceProcessEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName in~ ("chrome.exe", "msedge.exe", "teams.exe", "outlook.exe",
"npm.cmd", "node.exe", "w3wp.exe")
| where FileName in~ ("powershell.exe", "pwsh.exe", "cmd.exe", "wscript.exe", "cscript.exe",

```

```
"rundll32.exe", "regsvr32.exe")
| summarize count(), min(Timestamp), max(Timestamp) by DeviceName, InitiatingProcessFileName, FileName,
AccountName, ProcessCommandLine
| order by count_ desc
```

10. Threat Hunting

10.1. Hunt Hypothesis

If the reported activity is present locally, telemetry should show anomalous identity behavior, suspicious trusted-process child execution, abnormal network destinations, or unusual package/script activity around the initial access window.

10.2. Hunt Query

```
SigninLogs
| where TimeGenerated > ago(14d)
| where ResultType != 0 or RiskLevelAggregated in ("medium", "high") or ConditionalAccessStatus ==
"failure"
| summarize Events=count(), FirstSeen=min(TimeGenerated), LastSeen=max(TimeGenerated) by
UserPrincipalName, IPAddress, AppDisplayName, RiskLevelAggregated, ConditionalAccessStatus
| order by Events desc

DeviceNetworkEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName in~ ("chrome.exe", "msedge.exe", "teams.exe", "node.exe", "npm.exe",
"python.exe", "powershell.exe")
| where RemoteUrl !endswith ".microsoft.com" and RemoteUrl !endswith ".windows.com"
| summarize Connections=count(), FirstSeen=min(Timestamp), LastSeen=max(Timestamp) by DeviceName,
InitiatingProcessFileName, RemoteUrl, RemoteIP
| order by Connections desc
```

11. Risk Assessment

Dimension	Assessment
Likelihood	Medium priority based on current reporting, sector exposure, and adversary utility.
Impact	Potential business disruption, credential theft, data exposure, or operational degradation depending on local dependency.
Confidence	High for prioritization; local exposure requires validation.

12. Recommended Actions

Priority	Owner	Action
Critical	SOC	Run hunts across the prior 14 days and open cases for confirmed local hits.
High	Engineering	Validate patch, identity, package, and third-party controls relevant to the affected exposure path.
High	Incident response	Prepare containment steps and preserve logs for high-value systems.
Medium	Governance/vendor management	Request third-party evidence for aviation, supply-chain, identity, and managed-service dependencies.

13. References

- [1] CISA ICSA-26-197-03 NASA cFS HS advisory: <https://www.cisa.gov/news-events/ics-advisories/icsa-26-197-03>
- [2] CISA ICS advisories index: <https://www.cisa.gov/news-events/ics-advisories>
- [3] NASA Core Flight System project: <https://github.com/nasa/cFS>